Palm OS Programming Recipes

Using WinPaintBitmap Instead of WinDrawBitmap

A previous recipe (Drawing a Bitmap Family Member) described how to draw a bitmap family member using `WinDrawBitmap`. This recipe describes a similar method using `WinPaintBitmap` rather than `WinDrawBitmap`.

The advantage of using `WinPaintBitmap` is that it can draw to the screen using various draw modes; `WinDrawBitmap` uses only `winPaint` mode (also called "copy mode"). For more information about the draw modes available, read the definition of `WinDrawOperation` in *Palm OS Programmer's API Reference*.

Note that `WinPaintBitmap` was introduced in Palm OS 3.5. If you require `WinPaintBitmap`'s functionality, you will need to limit your application to handhelds running Palm OS 3.5 and higher.

1. Use Constructor for Palm OS to open the form that will contain the bitmap.

2. In the Project Window, create a new Bitmap Family.

3. Open the Bitmap Family resource and set the values for the properties:

    - Note that every member of the Bitmap Family must have the same Width and Height.
    - You can set one of the colors to be transparent when the bitmap is drawn to the screen.
    - You can add bitmaps of different depths to the bitmap family by selecting **Edit > New Family Member** (also by using Ctrl-K on Windows or Cmd-K on Macintosh). You can use up to five bitmap depths in one bitmap family, but bitmap images must be stored in increasing order of bit depth, and the bit depths must be unique. For backward compatibility, you should always have a 1-bit black and white bitmap as the first bitmap in the bitmap family.

4. Do not add a Form Bitmap to the form. Rather, use the following code to draw the bitmap family member:

```
//assuming the Bitmap Family was named Test
MemHandle hRsc = DmGetResource(bitmapRsc, TestBitmapFamily);
ErrFatalDisplayIf(!hRsc, "Could not get bitmap family resource");
BitmapType* bitmapP = (BitmapType*) MemHandleLock(hRsc);

//set the draw mode to winPaint and save the
//old draw mode to restore it later
WinDrawOperation oldDraMode = WinSetDrawMode(winMask);

WinPaintBitmap(bitmapP, topLeftCoordX, topLeftCoordY);

//set the draw mode back
WinSetDrawMode(oldDrawMode);
```

```
MemHandleUnlock(hRsc);
DmReleaseResource(hRsc);
```

Note that this code uses `winMask` mode ("`AND NOT`" mode). Any of the other draw modes can be used by substituting them for `winMask`.

To improve the performance of drawing the bitmap, you can call the "Get" and "Lock" functions well in advance to calling the "Draw" function. In addition, if you plan to reuse the same bitmap, you can call the "Unlock" and "Release" functions later in your code. For example, you could load and lock a number of bitmaps in an `AppStart` function, and then unlock and release them in an `AppStop` function.

The following related recipes describe alternative methods for drawing bitmaps to the screen:

- Adding a Bitmap Family to a Form
- Drawing a Bitmap Family Member
- Copying a Bitmap from an Offscreen Window